Week 13 web page: delivery guide

Instructor notes for Week13_HierarchicalModelsAndGAMs.html

General

- Page: https://kevintshoemaker.github.io/NRES-746/Week13_HierarchicalModelsAndGAMs.html. Script: `CODE_Week13_HierarchicalModelsAndGAMs.R` (same seeds, same numbers; each H1's section runs on its own after the first few lines).
- Monday, Nov. 16: *Formula to Model* review (due that day), the *Partial Pooling by Hand* WOD, then Bayesian hierarchical models (moved from the Week 12 page, which keeps a pointer).
- Tuesday, Nov. 17: final-project check-ins (30 min per group). No lab lecture, Act, or key.
- Wednesday, Nov. 18: splines, GAMs, and Gaussian processes. No in-class WOD; hand out the take-home *Spline by Hand* (due Mon., Nov. 23) at the end. Keys for both WODs: `wod/answer_keys/`.
- Thanksgiving: no class Wed., Nov. 25, so spatial models get one lecture and Lab 10. The GP section is the bridge; if it isn't reached Wednesday, open Nov. 23 with it. Going over *Spline by Hand* on Nov. 23 (~5 min) competes with that lecture; consider doing it during the Lab 10 opening instead.
- Data, all simulated (say so):
  - Drought sensitivity (seed 12), unchanged from Week 12: 12 stands, 194 stand-years; truth $\mu_a = 0.9$, $\mu_b = 0.25$, $\sigma_a = 0.15$, $\sigma_b = 0.10$, $\rho_{ab} = -0.6$, $\sigma_e = 0.12$.
  - Bd thermal gradient plate (seed 13): 48 cultures, 4-29 °C; truth a two-piece curve (Gaussian rise, quadratic fall), peak 0.30/day at 20 °C, zero at 28 °C, noise SD 0.03.
  - Take-home: reservoir chlorophyll, `data/ReservoirChlorophyll.csv` (44 of 52 weeks; spring bloom near week 15, small fall bloom near week 40). Push it to `gh-pages` and check the URL before handing out the WOD.
- Questions on the page have no answers. *Land:* is where discussion should end up.

Part 1: Bayesian hierarchical models (Monday, ~50 min)

Coding is de-emphasized this week: the hand-written Stan program is a page bonus, and the `brms` fit can be shown on the projector. Students who want to follow along need `brms` and `lme4` (section 6 of the script runs in about 1 min including the compile).

Notation change (Nov. 14), say it once at the start. The Week 11-13 pages now write

the AR(1) coefficient as ϕ (was ρ; matches Hyndman, `nlme`'s `Phi`, and Yang et al. 2026), the residual SD as σ_e, the random intercept as a_j with SD σ_a (was b_j, σ_b), the random slope SD as σ_b (was τ_b), and the intercept-slope correlation as ρ_{ab}. Plain ρ now means any correlation.

Formula to Model review (~8 min)

- Go over the count table (Part 1a) and one formula-to-equations item. Key in `wod/answer_keys/`.

What makes a model hierarchical? (~6 min)

- Layers: data given something unobserved, plus a model for the unobserved thing (stand effects, a spatial surface, true states in capture-recapture).
- A prior alone isn't a layer: Normal(0, 1) on each slope (no pooling) vs. an estimated σ_b (partial pooling).
- A marginal likelihood (one MVN, one negative binomial) can still be hierarchical; the label goes with the claim, not the likelihood.

WOD: Partial Pooling by Hand (~10 min real; 5 alone, then neighbor)

- Hand it out right after section 1. The handout gives the formula; the derivation is a page bonus (`#pooling-formula`), not covered in class.
- $w_A = 0.5$, $w_B = 0.8$; $\hat{a}_A = 3.6$, $\hat{a}_B = 3.8$. The ranking flips: pond A's raw mean is higher (4.2 vs. 4.0), but 4 frogs give a sampling variance (0.25) as big as the among-pond variance (0.25).
- One-line board summary: w_j = signal variance / (signal + sampling variance).

Limits and shrinkage (~3 min)

- $n_j \to \infty$: no pooling; $\sigma_a \to 0$: complete pooling. Name it shrinkage; contrast with ridge (λ chosen vs. σ_a estimated).

Partial pooling (~4 min)

- Drought data (same as Week 12 random slopes); the no-pooling figure.
- Code check: the formula matches `lmer()` to three decimals. $\hat{\sigma}_a = 0.113$, $\hat{\sigma}_e = 0.229$, $\hat{\mu} = 0.861$; weights 0.88 (30 yr) down to 0.55 (5 yr).
- Widget (`partial_pooling.html`): $\sigma_a = 0$ (every stand on the dashed line; weights 0.00), then $\sigma_a = 0.4$ (weights 0.99 and 0.94, nearly no pooling).
- "Set σ_a to 0, then drag it to 0.4. Which two pooling choices are these? Then compare S10 and S12 ... Which moves more toward the overall mean, and why?" *Land:* complete pooling, then (nearly) no pooling. S12 (5 yr, $w = 0.55$) moves a larger share of its distance (45% vs. 34% for S10, $w = 0.66$); absolute moves are almost equal (0.080 vs. 0.078 mm) because S10 starts farther out.

Fitting hierarchical models by MCMC (~3 min)

- Pooling intercepts and slopes together: write G (σ_a, σ_b, ρ_{ab}); `lmer()` MLEs are in `ml`.

- ML integrates the stand effects out; MCMC samples them. Four bullets: no integral; uncertainty carries through; priors help with few groups (the singular four-stand fit from Week 12 Q3); the price.

Priors for a covariance matrix (~2 min; one sentence if behind)

- Board: $G = \text{diag}(\sigma)\,\Omega\,\text{diag}(\sigma)$; LKJ gives $p(\rho) \propto (1-\rho^2)^{\eta-1}$ for 2×2.
- Widget (`lkj_prior.html`), 2×2: $\eta = 1$ flat ($P(|\rho| > 0.5) = 0.50$); $\eta = 2$ 0.31; $\eta = 0.5$ U-shaped, 0.67. Ask for a prediction before switching to 10×10.
- "For a 10×10 correlation matrix with $\eta = 1$, how likely is any one correlation to exceed 0.5 in absolute value? What does 'uniform' mean for a matrix that large?" *Land:* about 0.10 (prior SD 0.30). LKJ(1) is uniform over valid matrices, and most valid big matrices have small correlations.

The random slope model in brms (~10 min; laptops optional)

- One formula plus five priors (Normal(1, 0.5) intercept, Normal(0, 0.5) slope, half-Normal(0, 0.3) on SDs and σ_e, LKJ(2)). Order in `c()` doesn't matter; a `coef`-level prior beats a class-wide one; duplicates are an error (all checked in R). The `Intercept` prior sits at mean `precip10` (0.13), not 0. First compile ~20 s. No divergences; max $\hat{R}$ 1.007.
- Table: everything matches `lmer()` except ρ_{ab}: MLE -0.57; posterior median -0.36, 90% interval -0.75 to 0.17; truth -0.6. SDs 0.149 and 0.104 (truth 0.15, 0.10).
- Shrinkage figure: S11 (5 yr) moves most (slope 0.39 to 0.31); S01 and S02 (30 yr) barely move. Truth check: in this data set the short-record stands don't improve (slope error 0.057 both ways; intercepts 0.044 to 0.053); over 500 simulations slope error falls 0.070 to 0.051. Shrinkage wins on average, not every time.
- "Why is the posterior for ρ_{ab} so much wider ... closer to 0 than the MLE? Which summary is more honest?" *Land:* 12 stand-level pairs, each estimated with error, so the likelihood for ρ_{ab} is flat. The median moves for two reasons: MLE $-0.57 \to -0.44$ even with a flat LKJ(1) prior (bounded at -1, so the posterior is skewed toward 0 and its median sits off the peak; it also averages over the among-stand SDs), then $\to -0.36$ with LKJ(2) (prior pulls toward 0). The posterior is more honest: negative is likely, 0 isn't ruled out (-0.75 to 0.17).
- "Change `lkj(2)` to `lkj(1)`, then `lkj(10)` ..." *Land:* $\eta = 1$: median -0.44 (90% -0.81 to 0.17); $\eta = 10$: -0.14 (-0.44 to 0.21). Stand slopes move by at most 0.007 and 0.022. Split the room (each refit recompiles, ~20 s).
- The hand-written Stan program is now a bonus (`#stan-model`); point to it for anyone curious about the Cholesky factor and non-centering.

Discussion (if time; otherwise assign Q3 for project check-ins)

- Q1 (12 more stands or 20 more years?) *Land:* more stands; ρ_{ab} and the among-stand SDs describe the population of stands, and longer records can't add stands.
- Q2 (brms vs. your own Stan) *Land:* brms for anything a formula can say; your own Stan for custom process or observation models.

- Q3 (your project's groups) *Land:* open; ask for the grouping factor, its number of levels, and whether slopes plausibly vary. Good prep for Tuesday's check-ins.

Skip if behind: the LKJ widget (state the prior in one sentence), the η experiment, the discussion. The ridge-as-prior bonus is now needed Wednesday; mention it in one sentence at the end.

Part 2: Splines, GAMs, and Gaussian processes (Wednesday, ~50 min)

No laptops needed; all code is on the page.

Big picture and the bell curve (~6 min)

- MODELS box: when no bestiary function has the right shape, build one. Management question: above what temperature does Bd stop growing?
- Bell fit: $r_{\max} = 0.314$, $T_{\text{opt}} = 19.4$, width 4.72, $\sigma_e = 0.038$.
- "Which features of the scatter does the bell curve get wrong? Which matter for the management question?" *Land:* symmetric (the drop above the optimum is steeper than the rise), can't go below 0 (cultures die above ~28 °C), overpredicts at 27-29 °C. The upper tail is the one the question is about.

Splines are linear models (~12 min)

- Board: $f(x) = \sum \beta_k B_k(x)$; each B_k is a column of X. Show `head(bs(...))`: most entries 0 (local bumps). Knots at 10.4, 16.4, 22.65 °C for 6 functions.
- Figure: 6 functions fit well; 25 chase single cultures. Callback to the Week 10 connect-the-dots WOD.
- Widget (`spline_basis.html`), stops in order:
  1. B-spline, degree 1, 4 knots, "show each weighted basis function": tents; the sum bends only at knots. Ask where it can bend first.
  2. Degree 3: smooth joins (the "Where pieces join" card).
  3. Polynomial, highest power 12, randomize 3-4 times: most of the motion is at the edges (powers blow up near ± 1).
  4. Back to B-spline, 12 knots, independent weights: jagged. Switch to random walk and drag the step SD from 1 down to 0.05: smooth.
- Q1 (degree 1 vs. 3; polynomial) *Land:* degree 1 bends only at knots (kinks); cubic joins are smooth. Polynomials move everywhere when one weight changes and swing hardest at the ends.
- Q2 (random walk) *Land:* smooth curves come from weights that change slowly from neighbor to neighbor; that assumption is the penalty.

Penalizing wiggliness (~8 min)

- Board: penalized NLL with second differences; $\lambda = 0$ is `lm()`, $\lambda \to \infty$ is a straight line; edf from K down to 2.

- Penalty = prior: $\lambda = 1/(2\sigma_d^2)$, so the wiggly part is a random effect and REML picks λ the way `lmer()` picks σ_a. Callback to Monday's ridge bonus.
- Widget (`penalized_fit.html`), stops: $K = 40$, $\log_{10}\lambda = -4$ (edf 38.9, wiggly); drag up to 5 (edf 2.06, a sloped line). "Jump to the REML minimum": $\log_{10}\lambda = 0.9$, edf 9.1. Then $K = 20$ (minimum at 0.0, edf 8.1) and $K = 10$ (-1.2, edf 6.7). Tick "show the curve the data were simulated from" last.
- Q (edf and the sloped line; how much K matters) *Land:* edf falls to 2; the second-difference penalty is zero for any straight line, so the limit is the least-squares line, which slopes up (0.006 per °C). Once REML picks λ, $K = 10$, 20, 40 give edf 6.7-9.1 and nearly the same curve: K only needs to be big enough.

Fitting a GAM (~4 min)

- `gam()`: edf 8.2; `gam.vcomp()` prints the smooth's SD (0.017). Default GCV picks a wigglier curve (edf 13.9); always ask for REML.
- AIC table: bell -170.6, 6-function `lm()` -199.5, GAM -201.2 (ΔAIC 30.6 for the bell). GAM peak 20.2 °C, zero at 28.0 °C (truth 20 and 28); bell peak 19.4, never reaches 0.

Gaussian processes (~10 min; first thing to cut, moves to Nov. 23)

- Board: MVN on f itself; kernel $\alpha^2 \exp(-d^2/2\ell^2)$. Exponential kernel = AR(1) with $\phi = e^{-1/\ell}$ (point at the Week 12 V heatmap).
- Fit by `optim()`: $m = -0.057$, $\alpha = 0.32$, $\ell = 10.1$ °C, $\sigma_e = 0.027$. GP and GAM curves differ by at most 0.011. Don't dwell on m (it trades off with α when ℓ is long).
- Widget (`gp_draws.html`), stops: no points, squared exponential, $\ell = 1.5$ (correlation 0.80 at distance 1; falls to 0.1 at 3.2); "New random functions" twice. Add example points. Switch to exponential (0.51 at distance 1): rough draws, posterior band pinches at each point. Back to squared exponential, $\ell = 0.3$: between and beyond the points the posterior returns to the prior mean (0) and the band widens. Ask for a prediction before each switch.
- Q *Land:* exponential curves are continuous but rough (like an AR(1) series); squared exponential ones are smooth. With a short ℓ the data only inform nearby x; far away you get the prior back.
- Close: replace temperature with distance between sites, and this is a spatial model (kriging).

Discussion (~5 min; pick one)

- Q1 (32 °C: bell 0.009, GAM -0.315, GP -0.265) *Land:* all three are extrapolating. The bell can't go negative; the GAM and GP continue the last slope. None should be used; measure at 30-34 °C.
- Q2 (GAM beats bell by 30.6) *Land:* the bell is falsified as a description; asymmetry is corroborated only against this one rival. Add an asymmetric parametric rival (Briere, Sharpe-Schoolfield, or the two-piece Deutsch et al. 2008 form that generated the data).
- Q3 (manager wants one number) *Land:* report $\mathrm{CT}_{\max}$ from the GAM with a simulation interval (bonus: median 28.0, 95% 27.7-28.4; T_{opt} 20.2, 18.5-21.6), or fit an asymmetric

parametric curve with $\mathrm{CT}_{\max}$ as a parameter.

Last 2 min: hand out *Spline by Hand* (due Mon., Nov. 23). Take-home, laptops, AI labels on each coding item. Numbers they should get: AIC -5.9, -16.4, -41.8, -39.6 for $K = 4$, 8, 12, 20; `pnll` check 43.72; $\lambda = 10$ is closest to `gam()` (edf 9.7, peak week 15.5). Don't show the penalized NLL code before it's due; the page uses `gam()` only.

Bonuses (not planned for class): derived-quantity intervals; the `brms` smooth (`sds(stemp_1)` 0.53, on a different scale from `gam.vcomp()`); a table of GAM formula forms for final projects (Pedersen et al. 2019).